

SCDMS API Documentation

System: Submission & Commission Decision Management System (SCDMS) / Commission Decision Portal (CDP)

API version: 1.0.0

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1. Overview

The SCDMS REST API is **fully developed** and serves:

- The Commission Decision Portal (web UI)
- OPSC internal workflows (secretariat, units, compliance integration)
- **Machine-to-machine integration** (CMS today; **HRMIS and other partners** via agreed APIs)

Base URL (production): `https://<domain>/api/` (*assigned after DCDT virtual server and domain provisioning*)

Local / staging examples:

Environment	API base
Docker Compose (nginx)	<code>http://localhost:8080/api/</code>
Django dev server	<code>http://localhost:8000/api/</code>

Service discovery:

Resource	Path
API root (JSON)	GET /
Health check	GET /health/
OpenAPI 3 schema	GET /api/schema/
Swagger UI	GET /api/docs/
ReDoc	GET /api/redoc/
Endpoint inventory (auth)	GET /api/auth/api-inventory/

All JSON request and response bodies use **Content-Type: application/json** unless noted (file uploads use `multipart/form-data`).

2. Security and compliance

- Built with **Django 5** and **Django REST Framework**.
- Aligned with **Vanuatu's National Cyber Security Strategy 2030** (session limits, logout, TLS, secure headers).
- **JWT** access tokens (short-lived) and refresh rotation with blacklist on logout.

- Optional **TOTP 2FA** for interactive users (enforced per deployment policy).
- **API keys** for server-to-server integration (**X-API-Key**), inheriting the linked user's role permissions.
- **Role-based access control (RBAC)** on every submission and resource.
- **Audit logging** for create, update, workflow transitions, and downloads.
- **Rate limiting** on authentication and sensitive operations (see §5).

Technical Assessment Note (TAN) for DCDT hosting is in progress; production credentials and TLS are issued after server provisioning.

3. Authentication

3.1 JWT (interactive users and user-backed automation)

Obtain tokens

```
POST /api/auth/token/
Content-Type: application/json

{
  "username": "ministry.hr",
  "password": "<password>"
}
```

Response (200)

```
{
  "access": "<jwt-access>",
  "refresh": "<jwt-refresh>"
}
```

Setting	Default
Access token lifetime	30 minutes (JWT_ACCESS_MINUTES)
Refresh token lifetime	7 days (JWT_REFRESH_DAYS)
Refresh rotation	Enabled; old refresh blacklisted on use

Use access token

```
Authorization: Bearer <access>
```

Refresh

```
POST /api/auth/token/refresh/
{ "refresh": "<jwt-refresh>" }
```

Logout (blacklist refresh)

```
POST /api/auth/logout/
Authorization: Bearer <access>
{ "refresh": "<jwt-refresh>" }
```

Current user

```
GET /api/me/
Authorization: Bearer <access>
```

Returns username, role, `ministry_id`, `department_id`, and related profile fields used by the portal.

3.2 API key (recommended for HRMIS / CMS / batch jobs)

Server integrations should use a dedicated **service account** and API key created in SCDMS Admin (`/api/api-keys/`, `psc_admin` only).

```
X-API-Key: psc_<secret>
```

The key is bound to a Django user; **all RBAC rules apply** as that user. Keys are stored hashed; the plain key is shown once at creation.

You may send **either** `Authorization: Bearer ...` **or** `X-API-Key`, not both required.

3.3 Webhook authentication (CMS only)

Inbound webhooks do **not** use JWT. They require:

```
X-CMS-Callback-Key: <CMS_CALLBACK_SECRET>
```

Configured in environment on both CMS and SCDMS. See §10.

4. Common conventions

4.1 Pagination

List endpoints use page number pagination:

```
{
  "count": 120,
  "next": "https://host/api/submissions/?page=2",
  "previous": null,
  "results": [ ... ]
}
```

Default **page size**: 50 (`PAGE_SIZE`).

4.2 Errors

DRF standard shape:

```
{
  "detail": "Human-readable message."
}
```

Validation errors:

```
{
  "field_name": ["Error message."]
}
```

HTTP status codes: 400 validation, 401 unauthenticated, 403 forbidden, 404 not found, 429 throttled, 500 server error.

4.3 Timestamps

ISO 8601 with timezone (UTC in storage), e.g. 2026-05-22T04:30:00+11:00.

4.4 Reference numbers

Submissions receive a unique **reference_number**: PSC-YYYY-NNNNN (assigned on create).

5. Rate limits (indicative)

Scope	Limit
Anonymous	300/min
Authenticated user	600/min
Submission create	10/hour per user
OTP / password reset	5-10/min

Integrations should use **backoff** on 429 and idempotent retries where safe.

6. Core resources for partner integration

6.1 Ministries and departments

Method	Path	Description
GET	/api/ministries/	List ministries (line ministries + OPSC)
GET	/api/ministries/{id}/	Ministry detail

Method	Path	Description
GET	/api/departments/?ministry={id}	Departments for a ministry

Used to populate submission routing and filters. **Not** a full establishment register — see scope document.

6.2 Form metadata

Method	Path	Description
GET	/api/form-categories/PSC	business categories
GET	/api/form-types/?active=PSC	only PSC form types (code, name, is_digitized, category)
GET	/api/form-types/{id}	Single form type
GET	/api/form-fields/?form_type={id}	Field definitions for a form type

6.3 Submissions (primary integration object)

Method	Path	Description
GET	/api/submissions/	List (scoped by caller role)
POST	/api/submissions/	Create
GET	/api/submissions/{id}	Detail + workflow events + flags
PATCH	/api/submissions/{id}	Update metadata (role-dependent)
POST	/api/submissions/{id}	Advance workflow stage
GET	/api/submissions/{id}/events	Events
GET/POST	/api/submissions/{id}/documents	Documents (multipart on POST)
GET/POST/PUT	/api/submissions/{id}/digital-form/	Digital form answers (JSON data by field_key)

Resource identifier {id} (path parameter) The {id} segment in submission URLs is the **integer database primary key** (e.g. 901). It is **not** the business reference number.

Field	Type	Use
id (URL {id})	Integer PK	All /api/submissions/{id}/... routes
reference_number	String (e.g. PSC-2026-00142)	Display, search, CMS/HRMIS payloads
cdp_submission_pk	Integer	Same as id in webhook responses
cdp_submission_id	String	Same as reference_number in webhook responses

To resolve a submission by reference: GET /api/submissions/?search=PSC-2026-00142 (or list filters documented in OpenAPI), then call detail routes with the returned id.

Dynamic form validation (POST/PUT) data is validated server-side against the active **Form Builder** definitions for the submission's `form_type_code` (or the `form_type` id on first save). Unknown keys and missing required fields return **400 Bad Request**:

```
{
  "dynamic_form": [
    "Field 'salary_step_code' does not exist for Form PSC 3.6."
  ]
}
```

Integrators should load allowed keys from `GET /api/form-fields/?form_type={id}` before posting. Section-header rows (`field_type: section_header`) are layout-only and are not stored in `data`.

List filters (query parameters): vary by role; common filters include `current_stage`, `ministry`, `form_type_code`, search on title/reference. Use OpenAPI schema for the full list.

```
POST /api/submissions/
Authorization: Bearer <access>
Content-Type: application/json

{
  "title": "Appointment of Director Finance",
  "form_category": 5,
  "form_type_code": "PSC 3.6",
  "ministry": 3,
  "department": 12,
  "received_at": "2026-05-22T08:00:00+11:00",
  "notes": "Optional context"
}
```

Create submission — ministry / Commission track Response includes `id`, `reference_number`, `current_stage` (typically draft), and read-only flags.

Server-side rules: `ministry` and `department` may be inferred from the user profile for `ministry_hr` / `dept_admin`. `form_type_code` is normalized (e.g. PSC4.4 → PSC 4.4).

```
POST /api/submissions/
{
  "title": "Overseas training - Fiji",
  "form_type_code": "PSC 4.5",
  "received_at": "2026-05-22T08:00:00+11:00",
  "travel_endorsers": {
    "hod": 42,
    "director": 55,
    "dg": 7
  },
  "notes": ""
}
```

Create submission — Secretary-only travel (4.5 / 4.6) Response sets `secretary_only: true`, `requires_travel_letter` (true for 4.5 and 4.6).

Form 4.4 — domestic travel allowance: **only head_of_agency** (department director or ministry DG). Staff 4.4 is **not** accepted via API. DG path: self-sign then Secretary; department director: DG endorsement then Secretary.

```
POST /api/submissions/{id}/transition/
{
  "new_stage": "submitted",
  "remarks": "Lodged by ministry HR.",
  "acknowledge_gaps": false
}
```

Workflow transition Allowed `new_stage` values depend on **current stage**, **role**, `is_internal`, and `secretary_only`. See §8.

Travel submissions in `draft` → `submitted` require **all ministry endorsements** signed via `/api/submissions/{id}/sign-tr`

Key read-only fields on detail

Field	Meaning
<code>reference_number</code>	Business key PSC-YYYY-NNNNN
<code>current_stage</code>	Workflow stage code
<code>form_type_code</code>	e.g. PSC 3.6, PSC 4.5
<code>secretary_only</code>	Secretary path (no Commission sitting)
<code>requires_travel_letter</code>	4.5 / 4.6 letter after Secretary approval
<code>travel_endorsers</code>	{ "dg": 7, "hod": 42, ... } user IDs
<code>implementation_status</code>	Post-decision implementation
<code>cms_case_id</code>	Linked CMS case (compliance)
<code>subway_map</code>	UI workflow map (optional for integrations)

6.4 Travel endorsement endpoints (secretary-only forms)

Method	Path	Description
GET	<code>/api/submissions/{id}/required-endorsements/</code>	Required endorsements, signed state
POST	<code>/api/submissions/{id}/sign-travel</code>	Sign travel (section/key, optional remarks, approved for Secretary)
GET	<code>/api/submissions/{id}/travel-approval-pdf/</code>	Generate approval PDF (when applicable)

6.5 Post-decision (read / task systems)

Method	Path	Description
GET/POST	<code>/api/commission-tasks/</code>	Tasks after Commission decision
GET	<code>/api/reports/stats/</code>	Aggregated reporting (privileged roles)

7. HRMIS integration — division of responsibility

Layer	Owner	Status
SCDMS REST API (this document)	OPSC / IPDU	Available — use OpenAPI + service account
HRMIS APIs (employee lookup, validation, outcome write-back)	HRMIS unit	To be implemented by HRMIS

SCDMS is ready to **call** HRMIS once endpoints are agreed. Below is the **proposed contract** for HRMIS to implement.

7.1 Proposed HRMIS endpoints (HRMIS → called by SCDMS)

Authentication: mutual TLS or OAuth2 client credentials / API key — to be agreed.

GET `/api/v1/employees/{employee_id}` Resolve employee for pre-fill.

Response (example)

```
{
  "employee_id": "EMP-12345",
  "payroll_number": "12345",
  "full_name": "Jane Kalo",
  "ministry_code": "MFEM",
  "department_code": "FIN",
  "position_title": "Senior Accountant",
  "position_number": "POS-99",
  "employment_status": "permanent",
  "is_active": true
}
```

GET `/api/v1/employees/{employee_id}/validate-for-psc` Optional composite check before submit.

Response

```
{
  "valid": true,
  "errors": []
}
```

POST `/api/v1/psc-decision-outcomes` Called when SCDMS marks implementation complete (event-driven).

Request (example)

```
{
  "psc_reference": "PSC-2026-00142",
  "form_type_code": "PSC 3.6",
  "outcome_type": "appointment",
}
```



```

"effective_date": "2026-06-01",
"employee_id": "EMP-12345",
"position_title": "Director Finance",
"grade_or_level": "G10",
"acting_end_date": null,
"implemented_at": "2026-06-15T10:00:00+11:00"
}

```

Response: 202 Accepted with HRMIS transaction id (or 200 with body if HRMIS prefers synchronous ack — to be agreed in MOU).

Interconnection SLA (SCDMS behaviour when implemented):

- SCDMS posts outcomes **asynchronously** (Celery queue), not inside the user's browser request.
- SCDMS expects HRMIS to return **HTTP 202 Accepted** within **3000 ms** when the outcome is accepted for processing.
- If HRMIS is unreachable or does not respond in time, SCDMS applies **exponential backoff** (up to **5** attempts) before marking the implementation task **Failed Interconnection** in the audit trail.
- HRMIS should treat **psc_reference** as the idempotent business key; duplicate posts with the same reference should not create duplicate HR actions.

7.2 Deep linking (no API)

From	URL pattern
HRMIS → SCDMS	<code>https://<scdms-host>/submissions/{id}</code> where {id} is the integer PK; or search UI/API by PSC-2026-00142
SCDMS → HRMIS	<code>https://<hrmis-host>/employees/{employee_id}</code> (<i>HRMIS-defined</i>)

7.3 SCDMS endpoints HRMIS will typically call

Use case	SCDMS call
Poll status by reference	GET /api/submissions/?search=PSC-2026-00142 or detail by id
Read outcome / stage	GET /api/submissions/{id}/ → <code>current_stage</code> , <code>implementation_status</code>
Register linked matter	Usually created in SCDMS by ministry HR; HRMIS does not duplicate create unless using service account

8. Workflow stages (reference)

Code	Label
<code>draft</code>	Draft
<code>submitted</code>	Submitted to PSC
<code>received_by_psc</code>	Received by PSC

Code	Label
returned_for_clarification	Returned for Clarification
registered_routed	Registered and Routed
manager_checklist_review	Manager Checklist Review
under_assessment	Under Assessment
compliance_under_review	Compliance Under Review (CMS)
forwarded_to_commission	Forwarded to Commission
commission_sitting	Commission Sitting
approved / rejected / returned	Commission outcomes
secretary_review	Secretary Review (internal / travel / CMS register)
minutes_drafted_signed	Minutes Drafted and Signed
decision_entered_assigned	Decision Entered and Assigned
under_implementation	Under Implementation
implementation_report	Implementation Report

Secretary-only graph (travel 4.4–4.6, simplified):

draft → submitted → secretary_review → approved / rejected / returned_for_clarification

9. CMS integration (reference)

Already live between **Case Management System (CMS)** and SCDMS.

Method	Path	Auth	Purpose
POST	/api/webhooks/cms-register	X-CMS-Callback-Key	Register compliance case → CDP submission
POST	/api/webhooks/cms-sign-off	X-CMS-Callback-Key	Compliance Manager sign-off → advance stage

Webhook security (operations):

- Rotate CMS_CALLBACK_KEY / X-CMS-Callback-Key via environment variables on a scheduled basis.
- Local and CI test suites should simulate production **TLS** and security headers required under the National Cyber Security Strategy 2030 (including self-signed certificates where DCDT endpoints are not yet available).
- **Planned v1.1.0:** upgrade CMS callbacks from shared secret header to **HMAC-SHA256** request signing for non-repudiation.

Register payload (example)

```
{
  "cms_case_id": "CMS-2026-0042",
  "cms_case_reference": "CMS-2026-0042",
  "title": "Disciplinary matter - Smith",
  "case_family": "employee_disciplinary",
  "form_type_code": "COMP-SMDR",
  "notes": "",
  "registered_by": "compliance.manager"
}
```

Register response (201)

```
{
  "cdp_submission_id": "PSC-2026-00142",
  "cdp_submission_pk": 901,
  "cdp_callback_url": "https://<host>/api/webhooks/cms-signoff/",
  "current_stage": "secretary_review",
  "cms_case_id": "CMS-2026-0042"
}
```

10. AI and async operations (optional)

Several endpoints queue **Celery** jobs (brief generation, duplicate detection, document OCR). Pattern:

```
POST /api/submissions/{id}/generate-brief/
GET  /api/submissions/{id}/      → poll ai_brief_processed, ai_brief_summary
```

Requires workers and API keys for Anthropic/OpenAI in deployment. Not required for HRMIS v1 integration.

11. Full endpoint catalogue

The authoritative machine-readable catalogue is:

1. **OpenAPI 3:** GET /api/schema/
2. **Swagger UI:** /api/docs/
3. **Authenticated inventory:** GET /api/auth/api-inventory/

The README in the repository root contains a **summary table** of major route groups; OpenAPI supersedes it where they differ.

12. IPDU team — direct action items

Action	Owner	Notes
DCDT server handshake	IPDU + DCDT	Run integration tests with strict TLS/headers (NCSS 2030) using local self-signed certs until production domain is live

Action	Owner	Notes
HRMIS MOU / RFC	IPDU + HRMIS technical unit	Use §7.1 as the formal contract template; agree auth, 202 SLA, and idempotent <code>psc_reference</code>
CMS webhook key rotation	IPDU ops	Rotate <code>X-CMS-Callback-Key</code> via env; plan HMAC signing in v1.1.0
Dynamic form contract tests	IPDU dev	Assert 400 <code>dynamic_form</code> errors in partner test packs for unknown <code>field_key</code> values

13. Changelog and contact

Version	Date	Notes
1.1	May 2026	Dynamic form validation; <code>{id}</code> vs <code>reference_number</code> ; HRMIS retry SLA; IPDU checklist; CMS webhook ops
1.0	May 2026	Initial partner-facing API guide; HRMIS proposed contract; Form 4.4 director/DG rules

For a **service account** and API key for HRMIS staging, contact IPDU after mutual agreement / security review.